

Spring Boot Basics - Exercises

Assignment 1: Create a Simple REST API

Objective: Build a Spring Boot application with a simple REST controller.

Requirements:

- Use Spring Initializr with `spring-boot-starter-web`.
- Create a `UserController` with endpoints:
 - `GET /users/{id}` – returns user by ID.
 - `POST /users` – adds a new user.
- Use `@RestController`, `@PathVariable`, `@RequestBody`

Assignment 2: Externalize Configuration

Objective: Learn `application.properties` or `application.yml` usage.

Requirements:

- Create a custom config property (e.g., `app.message=Welcome to Spring Boot!`)
- Inject it using `@Value("${app.message}")` in a controller.
- Print the message at an endpoint `/message`.

Assignment 3: Use Spring Boot Starters and Auto-Configuration

Objective: Understand Spring Boot starters and auto-configuration.

Requirements

- Use `spring-boot-starter-web` and `spring-boot-starter-thymeleaf`.
- Build a small web app that renders a greeting using Thymeleaf.
- Observe how no XML configuration or servlet setup is needed.

Assignment 4: Configure Beans Using Java Config

Objective: Learn how to define beans using @Configuration and @Bean.

Requirements:

- Create a GreetingService class.
- Create a configuration class that returns a GreetingService bean.
- Inject this service into a controller and expose /greet.

Assignment 5: Build a CRUD REST Service

Objective: Use REST + in-memory persistence (no DB).

Requirements:

- Create an entity class `Product` with fields: id, name, price.
- Store data in a `List<Product>` or `Map<Long, Product>`.
- Implement endpoints: add, list, update, delete.

Assignment 5: Compare Spring Boot with Spring (Research Task)

Objective: Understand differences between traditional Spring and Spring Boot.

Requirements:

- Compare manual vs auto configuration.
- Create a comparison table (deployment, setup, configuration, etc.).
- Present findings in a small document or short presentation.